

WASP-11b

R-0914

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_240829 · created 2026-07-29T07:01:33+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460551.8601 +/- 0.0035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.154 +/- 0.051
Transit depth (Rp/Rs)^2	2.36 +/- 1.57 [%]
Orbital Inclination (inc)	89.96 +/- 1.74
Airmass coefficient 1 (a1)	0.39 +/- 0.2
Airmass coefficient 2 (a2)	0.83 +/- 0.25
Scatter in the residuals of the lightcurve fit is	53.25 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	3.5
Transit Duration (day)	0.103 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.154 ± 0.051 vs 0.131 (deviation 0.45σ)
Duration fit vs published	0.103 d vs 0.1075 d (deviation 0.41σ)
Mid-time O–C	-7.0 min
Depth SNR	3.0
Residual scatter	53.25 %

Light curve

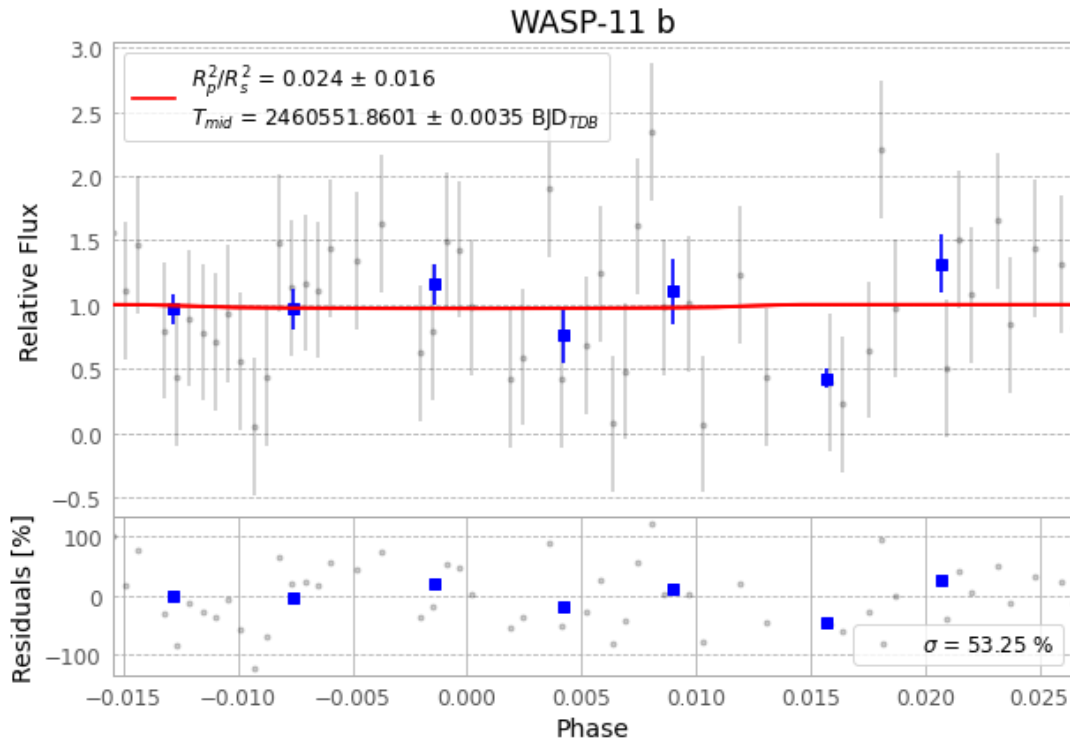


Figure 1 — FinalLightCurve_WASP-11 b_2024-08-29.png (SHA-256 d06f284a8fc456d7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [418, 434], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b41e9f68c0abbb7f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

77 FITS frames (51 MB), HATP-10240829071215.FITS → HATP-10240829110015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-240829.log (8316dcfd7f40...), FinalLightCurve_WASP-11 b_2024-08-29.png (d06f284a8fc4...), FinalLightCurve_WASP-11 b_2024-08-29.csv (2cb4d38a6b55...)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 07:01Z.